

EEE 554 Fall 2026

Homework 2: Questions and Study Solutions

Questions, worked steps, and final results

1. Binomial-to-Poisson Limit

Question. The binomial PMF is $P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}$ for $k = 0, 1, \dots, n$. Consider $n \rightarrow \infty$ and $p \rightarrow 0$, while $\lambda = np$ remains constant. Express the limit in terms of λ . What PMF is it?

1. Since $\lambda = np$, substitute $p = \lambda/n$.
2. For fixed k , the first factor approaches $\binom{n}{k} (\lambda/n)^k \rightarrow \lambda^k/k!$
3. The remaining factor approaches $(1 - \lambda/n)^{n-k} \rightarrow e^{-\lambda}$
4. Multiply the two limiting factors.

Final result.

$$P[X = k] \rightarrow e^{-\lambda} \lambda^k / k!, \quad k = 0, 1, 2, \dots$$

This is the Poisson(λ) PMF.

2. Parity Transformation of a Poisson RV

Question. Let X be Poisson with parameter λ . Define $Y = 1$ if X is even and $Y = -1$ if X is odd. Find the PMF of Y .

1. For $Y = 1$, sum the Poisson PMF over $X = 0, 2, 4, \dots$
2. For $Y = -1$, sum the PMF over $X = 1, 3, 5, \dots$
3. Use the even and odd exponential series, or equivalently use $e^\lambda = \cosh(\lambda) + \sinh(\lambda)$

Final result.

$$P[Y = 1] = (1 + e^{-2\lambda})/2$$

$$P[Y = -1] = (1 - e^{-2\lambda})/2$$

3. Best Constant Estimate of a Bernoulli RV

Question. Predict a Bernoulli random variable X with parameter p , where $0 \leq p \leq 1$, using its best constant estimate $\hat{X}$. (a) Find $\hat{X}$. (b) Find the mean-square error. (c) When is the error maximum, and why?

1. Let the constant estimate be a , and minimize $J(a) = E[(X - a)^2]$

2. Expanding and differentiating gives $dJ/da = -2E[X] + 2a = 0$
3. Therefore $a = E[X] = p$.
4. For a Bernoulli RV, $X^2 = X$, so its variance and minimum MSE are $p(1 - p)$.
5. The parabola $p(1 - p)$ is largest at $p = 1/2$.

Final result.

$$\hat{X} = p, \text{ MSE} = p(1 - p), \text{ MSE}_{\max} = 1/4 \text{ at } p = 1/2$$

4. Binomial Characteristic Function

Question. For a binomial RV X : (a) calculate its characteristic function; (b) determine its expected value and variance from the characteristic function; and (c) show it becomes the Poisson characteristic function as $n \rightarrow \infty$, $p \rightarrow 0$, with $\lambda = np$ fixed.

1. Start with $\Phi_X(\omega) = E[e^{j\omega X}] = \sum e^{j\omega k} P[X = k]$
2. Insert the binomial PMF and apply the binomial theorem.
3. Use $E[X] = \Phi'_X(0)/j$ and $E[X^2] = -\Phi''_X(0)$, then calculate variance.
4. Set $p = \lambda/n$ and use the exponential limit $(1 + z/n)^n \rightarrow e^z$

Final result.

$$\begin{aligned}\Phi_X(\omega) &= [1 - p + pe^{j\omega}]^n \\ E[X] &= np, \text{ Var}(X) = np(1 - p) \\ \Phi_X(\omega) &\rightarrow e^{\lambda(e^{j\omega} - 1)}\end{aligned}$$

5. Q-Function Identity

Question. For a PSK system, the probability of error uses the Q-function. If $N \sim N(0,1)$, prove $Q(x) = P[N > x]$.

1. Write the definition of the Q-function.
2. Write the standard normal PDF.
3. The integral of the PDF from x to infinity is, by definition, the upper-tail probability.

Final result.

$$Q(x) = (1/\sqrt{2\pi}) \int_x^\infty e^{-t^2/2} dt = P[N > x]$$

6. Affine Transformation of a Gaussian RV

Question. Let $X \sim N(0,1)$ and use the affine transformation $Y = 2X + \mu$. Prove $Y \sim N(\mu, 4)$.

1. Solve for X : $X = (Y - \mu)/2$
2. Apply the change-of-variables rule with the Jacobian magnitude $1/2$.
3. Substitute the standard normal PDF and simplify.
4. The coefficient of $(y - \mu)^2$ identifies variance 4.

Final result.

$$f_Y(y) = (1/\sqrt{8\pi})e^{-(y - \mu)^2/8}, \quad Y \sim N(\mu, 4)$$

7. Squared Standard Normal RV

Question. Let $X \sim N(0,1)$ and $Y = X^2$. Find $f_Y(y)$.

1. The support is $y \geq 0$.
2. For $y > 0$, solve $x^2 = y$ to get the two inverse values $x = \sqrt{y}$ and $x = -\sqrt{y}$.
3. Apply the change-of-variables rule to both branches; each has derivative magnitude $2\sqrt{y}$.
4. Add the two equal normal-density contributions.

Final result.

$$f_Y(y) = e^{-y/2}/\sqrt{2\pi y}, \quad y > 0$$
$$f_Y(y) = 0, \quad y \leq 0$$

8. Clipped Rayleigh RV

Question. Let X be Rayleigh. Define $Y = X$ for $0 \leq X < x_{\max}$ and $Y = x_{\max}$ for $X \geq x_{\max}$. Find the distribution of Y .

1. Below the clipping threshold, the transformation is the identity, so the continuous density remains the Rayleigh density.
2. Every value of X at or above $x_{\max}$ maps to one value of Y .
3. That tail probability becomes a discrete probability mass at $x_{\max}$.

Final result.

$$f_Y(y) = f_X(y), \quad 0 \leq y < x_{\max}$$
$$P[Y = x_{\max}] = 1 - F_X(x_{\max}) = e^{-x_{\max}^2/(2\sigma^2)}$$

9. Exponential Characteristic Function and n-th Moment

Question. Let X be exponential with rate λ . (a) Calculate $\Phi_X(\omega)$. (b) Calculate the n -th moment using the characteristic function.

1. Start with the exponential PDF $\lambda e^{-\lambda x} u(x)$.
2. Integrate $e^{j\omega x}$ times the PDF from 0 to infinity.
3. For moments, use $E[X^n] = \Phi_X^{(n)}(0) / j^n$
4. Differentiate the rational characteristic function n times and evaluate at zero.

Final result.

$$\Phi_X(\omega) = \lambda / (\lambda - j\omega)$$

$$E[X^n] = n! / \lambda^n$$

10. Rare Events and Importance Sampling

Question. Let $X \sim N(3, 1)$. (a) Express $P[X > 8]$ as an integral and estimate samples needed for 10 events. (b) For $f_Y(y) = e^{-(y-3)} u(y-3)$, find samples needed for 10 events $Y > 8$. (c) Express $P[X > 8]$ using importance sampling and give the explicit weighting function.

(a) Gaussian sampling

1. Standardize: $(8 - 3)/1 = 5$.
2. Therefore the upper-tail probability is $Q(5)$.
3. The expected number of exceedances in N trials is $NQ(5)$. Set this equal to 10.

$$P[X > 8] = Q(5) \approx 2.87 \times 10^{-7}$$

$$N \approx 10/Q(5) \approx 3.5 \times 10^7 \text{ samples}$$

(b) Shifted exponential comparison RV

1. For $Y > 8$, the excess over the shift is $8 - 3 = 5$.
2. The exponential tail is e^{-5} .
3. Set the expected number of successes $N e^{-5}$ equal to 10.

$$P[Y > 8] = e^{-5} \approx 6.74 \times 10^{-3}$$

$$N \approx 10/e^{-5} \approx 1.48 \times 10^3 \text{ samples}$$

(c) Importance sampling

1. Write the target probability as an integral using the target Gaussian density f_X .

2. Multiply and divide the integrand by the proposal density f_Y .
3. Recognize the resulting integral as an expectation under Y .

$$P[X > 8] = E_Y[I(Y > 8)f_X(Y)/f_Y(Y)]$$

$$h(y) = I(y > 8) \cdot (1/\sqrt{2\pi})e^{-(y-3)^2/2 + (y-3)}$$